

PROGRAMMING FUNDAMENTALS — GROUP TASK

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Programming Fundamentals – Assignment 01 (BBA – Business Analytics)

C++ Solutions (Q1 – Q11)

Q1. Sales Revenue Calculator (5 Marks)

```
#include <iostream>
#include <string>
using namespace std;

int main() {
    string productName;
    int quantity;
    double sellingPrice, revenue;

    cout << "Enter product name: ";
    getline(cin, productName);
    cout << "Enter quantity sold: ";
    cin >> quantity;
    cout << "Enter selling price per unit: ";
    cin >> sellingPrice;

    revenue = quantity * sellingPrice;

    cout << "Product: " << productName << endl;
    cout << "Quantity Sold: " << quantity << endl;
    cout << "Selling Price: Rs. " << sellingPrice << endl;
    cout << "Total Revenue: Rs. " << revenue << endl;

    return 0;
}
```

Sample Input:
Laptop Bag
25
2500

Sample Output:
Product: Laptop Bag
Quantity Sold: 25
Selling Price: Rs. 2500
Total Revenue: Rs. 62500

Explanation:
The program takes product name, quantity and price as input, calculates revenue (Quantity x Selling Price) and displays the result.

Q2. Student Business Profile (5 Marks)

```
#include <iostream>
#include <string>
using namespace std;

int main() {
    string name;
    int age, semester;
    double cgpa;
    bool isAnalytics;

    cout << "Enter student name: ";
    getline(cin, name);
    cout << "Enter age: ";
    cin >> age;
    cout << "Enter BBA semester: ";
    cin >> semester;
    cout << "Enter CGPA: ";
    cin >> cgpa;
    cout << "Is Business Analytics student (1=Yes, 0=No): ";
    cin >> isAnalytics;

    cout << "\n--- Student Profile ---\n";
    cout << "Name: " << name << endl;
    cout << "Age: " << age << endl;
    cout << "BBA Semester: " << semester << endl;
    cout << "CGPA: " << cgpa << endl;
    cout << "Business Analytics Student: " << (isAnalytics ? "Yes" : "No") << endl;

    return 0;
}
```

Sample Input:
Ali Khan
20
4
3.45
1

Sample Output:
--- Student Profile ---
Name: Ali Khan
Age: 20
BBA Semester: 4
CGPA: 3.45
Business Analytics Student: Yes

Explanation:
Uses string, int, double and bool variables and displays the information in a formatted way.

Q3. Business Profit Calculator (5 Marks)

```
#include <iostream>
using namespace std;

int main() {
    int units;
    double sellingPrice, costPerUnit, fixedCost;
    double revenue, variableCost, totalCost, profit;

    cout << "Units sold: ";
    cin >> units;
    cout << "Selling price per unit: ";
    cin >> sellingPrice;
    cout << "Cost per unit: ";
    cin >> costPerUnit;
    cout << "Fixed operating cost: ";
    cin >> fixedCost;

    revenue = units * sellingPrice;
    variableCost = units * costPerUnit;
    totalCost = variableCost + fixedCost;
    profit = revenue - totalCost;

    cout << "Total Revenue: Rs. " << revenue << endl;
    cout << "Variable Cost: Rs. " << variableCost << endl;
    cout << "Total Cost: Rs. " << totalCost << endl;
    cout << "Profit: Rs. " << profit << endl;

    return 0;
}
```

Example Input:
500
1200
750
100000

Sample Output:
Total Revenue: Rs. 600000
Variable Cost: Rs. 375000
Total Cost: Rs. 475000
Profit: Rs. 125000

Explanation:
Uses given formulas to calculate revenue, variable cost, total cost and profit.

Q4. Employee Salary Calculator (5 Marks)

```
#include <iostream>
using namespace std;

int main() {
    double basicSalary, houseAllowance, medicalAllowance, transportAllowance, tax, grossSalary, netSalary;

    cout << "Basic salary: ";
    cin >> basicSalary;
    cout << "House allowance: ";
    cin >> houseAllowance;
    cout << "Medical allowance: ";
    cin >> medicalAllowance;
    cout << "Transport allowance: ";
    cin >> transportAllowance;
    cout << "Tax: ";
    cin >> tax;

    grossSalary = basicSalary + houseAllowance + medicalAllowance + transportAllowance;
    netSalary = grossSalary - tax;

    cout << "Gross Salary: Rs. " << grossSalary << endl;
    cout << "Net Salary: Rs. " << netSalary << endl;

    return 0;
}
```

Sample Input:
50000
5000
3000
2000
5000

Sample Output:
Gross Salary: Rs. 66000
Net Salary: Rs. 61000

Explanation:
Adds allowances to get gross salary and subtracts tax to get net salary.

Q5. Discount and Final Price (5 Marks)

```
#include <iostream>
using namespace std;

int main() {
    double price, discountPercent;
    int quantity;
    double grossAmount, discountAmount, finalAmount;

    cout << "Product price: ";
    cin >> price;
    cout << "Quantity: ";
    cin >> quantity;
    cout << "Discount percentage: ";
    cin >> discountPercent;

    grossAmount = price * quantity;
    discountAmount = grossAmount * (discountPercent / 100);
    finalAmount = grossAmount - discountAmount;

    cout << "Gross Amount: Rs. " << grossAmount << endl;
    cout << "Discount: Rs. " << discountAmount << endl;
    cout << "Final Amount: Rs. " << finalAmount << endl;

    return 0;
}
```

Sample Input:
2000
5
10

Sample Output:
Gross Amount: Rs. 10000
Discount: Rs. 1000
Final Amount: Rs. 9000

Explanation:
Applies discount percentage on gross amount to get final price.

Q6. Sales Commission Calculator (5 Marks)

```
#include <iostream>
#include <string>
using namespace std;

int main() {
    string name;
    double salesAmount, commissionRate, commission;

    cout << "Salesperson name: ";
    getline(cin, name);
    cout << "Total sales amount: ";
    cin >> salesAmount;
    cout << "Commission rate (%): ";
    cin >> commissionRate;

    commission = salesAmount * (commissionRate / 100);

    cout << "Salesperson: " << name << endl;
    cout << "Total Sales: Rs. " << salesAmount << endl;
    cout << "Commission Rate: " << commissionRate << "% << endl;
    cout << "Commission Amount: Rs. " << commission << endl;

    return 0;
}
```

Example Input:
Ali
500000
5

Sample Output:
Salesperson: Ali
Total Sales: Rs. 500000
Commission Rate: 5%
Commission Amount: Rs. 25000

Explanation:
Calculates commission using Sales x (Rate/100).

Q7. Break-Even Calculation (5 Marks)

```
#include <iostream>
using namespace std;

int main() {
    double fixedCost, sellingPrice, variableCost, breakEvenUnits;

    cout << "Fixed cost: ";
    cin >> fixedCost;
    cout << "Selling price per unit: ";
    cin >> sellingPrice;
    cout << "Variable cost per unit: ";
    cin >> variableCost;

    breakEvenUnits = fixedCost / (sellingPrice - variableCost);

    cout << "Break Even Units: " << breakEvenUnits << endl;

    return 0;
}
```

Example Input:
100000
1200
750

Sample Output:
Break Even Units: 266.67

Explanation:
Uses formula FixedCost / (SP - VC) and double for accurate result.

Q8. Sales Target Evaluation (5 Marks)

```
#include <iostream>
#include <string>
using namespace std;

int main() {
    string name;
    double target, actual;

    cout << "Salesperson name: ";
    getline(cin, name);
    cout << "Sales target: ";
    cin >> target;
    cout << "Actual sales: ";
    cin >> actual;

    if (actual >= target) {
        cout << name << " has Achieved the Target" << endl;
    } else {
        cout << name << " has NOT Achieved the Target" << endl;
    }

    return 0;
}
```

Sample Input:
Sara
500000
450000

Sample Output:
Sara has NOT Achieved the Target

Explanation:
Compares actual sales with target using if-else.

Q9. Customer Discount Eligibility (5 Marks)

```
#include <iostream>
using namespace std;

int main() {
    double amount, discountRate, discountAmount;
    cin >> amount;

    cout << "Purchase amount: Rs. ";
    cin >> amount;

    if (amount >= 50000) discountRate = 15;
    else if (amount >= 20000) discountRate = 10;
    else if (amount >= 10000) discountRate = 5;
    else discountRate = 0;

    discountAmount = amount * (discountRate / 100);
    finalAmount = amount - discountAmount;

    cout << "Purchase Amount: Rs. " << amount << endl;
    cout << "Discount Rate: " << discountRate << "% << endl;
    cout << "Discount Amount: Rs. " << discountAmount << endl;
    cout << "Final Payable Amount: Rs. " << finalAmount << endl;

    return 0;
}
```

Sample Input:
60000

Sample Output:
Purchase Amount: Rs. 60000
Discount Rate: 10%
Discount Amount: Rs. 6000
Final Payable Amount: Rs. 54000

Explanation:
Uses if-else-if to determine discount based on purchase amount.

Q10. Employee Bonus & Performance (5 Marks)

```
#include <iostream>
#include <string>
using namespace std;

int main() {
    string name, classification;
    double basicSalary, bonus = 0, finalSalary;
    int score;

    cout << "Employee name: ";
    getline(cin, name);
    cout << "Basic salary: ";
    cin >> basicSalary;
    cout << "Performance score: ";
    cin >> score;

    if (score >= 80 && score <= 100) {
        // Excellent
        bonus = 0.15 * basicSalary;
    } else if (score >= 70) {
        // Good
        bonus = 0.10 * basicSalary;
    } else {
        // Needs Improvement
        bonus = 0;
    }

    finalSalary = basicSalary + bonus;

    cout << "Name: " << name << endl;
    cout << "Classification: " << classification << endl;
    cout << "Basic Salary: Rs. " << basicSalary << endl;
    cout << "Final Salary: Rs. " << finalSalary << endl;

    return 0;
}
```

Sample Input:
Aliza
60000
92

Sample Output:
Name: Aliza
Classification: Excellent
Bonus: Rs. 12000
Final Salary: Rs. 72000

Explanation:
Uses score ranges to give classification and bonus percentage.

Q11. Billing System (5 Marks)

```
#include <iostream>
#include <string>
using namespace std;

int main() {
    string item;
    int quantity;
    double price, taxRate, discountRate, subtotal, tax, discount, total;

    cout << "Item name: ";
    getline(cin, item);
    cout << "Price per item: Rs. ";
    cin >> price;
    cout << "Tax rate (%): ";
    cin >> taxRate;
    cout << "Discount rate (%): ";
    cin >> discountRate;

    subtotal = quantity * price;
    tax = subtotal * (taxRate / 100);
    discount = subtotal * (discountRate / 100);
    total = subtotal + tax - discount;

    cout << "Fixed or predetermined" << endl;
    cout << "Item: " << item << endl;
    cout << "Quantity: " << quantity << endl;
    cout << "Subtotal: Rs. " << subtotal << endl;
    cout << "Tax: Rs. " << tax << endl;
    cout << "Discount: Rs. " << discount << endl;
    cout << "Total Amount: Rs. " << total << endl;

    return 0;
}
```

Sample Input:
Notebook
3
1500
5
10

Sample Output:
Item: Notebook
Quantity: 3
Subtotal: Rs. 4500.00
Tax: Rs. 225.00
Discount: Rs. 450.00
Total Amount: Rs. 4275.00

Explanation:
Calculates subtotal, tax, discount and final total amount.